



Assessing the Effectiveness of State Regulation and Enforcement of Hazardous Waste^{*}

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Abstract

This paper assesses the effectiveness of different state hazardous waste regulations and policies in promoting compliance using data for 8000 facilities across the United States. Because data on compliance is available only if a facility is inspected, the analysis uses a censored bivariate probit. Adoption of voluntary pollution prevention programs appears to decrease violations in general while adoption of strict liability and allocating a higher percentage of employees to regional offices appear to decrease Class 1 violations only. Although spending on waste programs also appears to decrease Class 1 violations, spending is positively correlated with the overall probability of violation.

This paper examines the effect that state hazardous waste regulations and policies have on firm compliance with those regulations. As with all environmental regulations, the United States Environmental Protection Agency (EPA) has established the minimum level (the federal level) of hazardous waste regulation. However, states can and do make additional regulations concerning hazardous waste. Moreover, the intent of the Resource Conservation and Recovery Act (RCRA) is that states assume primary responsibility for implementing hazardous waste regulations. Thus, hazardous waste regulation, implementation, and enforcement vary across states. The purpose of this paper is to assess the effectiveness of different state hazardous waste regulations and policies in promoting facility compliance. The remainder of the paper is organized as follows. Section 1 provides a brief description of hazardous waste regulation and enforcement and a review of the empirical literature on hazardous waste compliance. Section 2 describes the theoretical framework and empirical approach of this analysis. Section 3 discusses the data used in the analysis and the results, and section 4 concludes.

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1. Background

Under Subtitle C of RCRA, EPA established minimum standards with which all facilities that generate, store, transport, treat, or dispose hazardous waste must comply.¹ States may adopt the federal regulations or can establish their own regulations. EPA will authorize a state to run its own hazardous waste program if the state's regulations are at least as stringent as the federal regulations and the state has demonstrated how it will enforce the program, disseminate information to the public, and fund the program's operation. If a state's program has not been authorized, EPA implements and enforces the federal regulations. Hazardous waste regulations are enforced primarily through facility inspections, although violations which result in damage to human health or the environment are sometimes self-reported or reported by third parties. Once a violation is discovered, the enforcement agency can initiate administrative, civil, or criminal actions, all of which can include penalty provisions.

Although a number of papers examine compliance with environmental regulations for media such as air and water, little has been written on hazardous waste compliance.² One study by Hamilton (1996) evaluates the effect of informal and formal rules in enforcing RCRA regulations. This paper focuses on the federal regulator's decision-making process and finds that informal rules (i.e., guidance and policy documents) increase regulatory stringency and the penalties associated with violations. Davis and Davis (1999) also focus on the regulator's decision, but examine state efforts rather than federal. Using aggregate state data, the authors analyze the determinants of state enforcement actions and find that economic dependence on industries that produce hazardous waste, the political ideology of state officials, and citizen membership in environmental groups significantly influence enforcement efforts.

Only one paper examines the firm's compliance decision. Stafford (2002) considers the effect of a federal increase in the penalties associated with violations of hazardous waste regulations on firm compliance. The results indicate that violations decreased in response to EPA's 1992 penalty increase. Additionally, although variation in state regulations and policies is not emphasized, the results suggest that there is significant variation in compliance across states. This paper expands on those results by examining more explicitly the effects of different state regulations and policies on facility hazardous waste compliance. The analysis uses data on over 8,000 facilities located across the United States and includes significant variation across states.

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- 1 Environmental damages from hazardous waste at inactive or abandoned sites (as opposed to ongoing operations) as well as hazardous waste resulting from spills that require an emergency response are covered by the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) of 1980, commonly referred to as Superfund, rather than by RCRA.
 - 2 Heyes (2000) provides an excellent survey of both the theoretical and empirical literature on environmental compliance.

2. Theoretical Framework and Empirical Approach

Assuming rational behavior, facilities will violate environmental regulations if the expected benefit from violation, i.e., the avoided cost of compliance, is greater than the expected cost of violation. The expected cost depends in part on the penalty that will be assessed if the violation is detected as well as the probability of detection. The cost of compliance is obviously facility specific as it depends on the type and quantity of waste the facility generates. The probability of inspection is also facility specific because inspectors target facilities based on potential risks to human health and the environment as well as past compliance history. Finally, the penalty if the firm is caught is facility specific, as the RCRA penalty policy takes the gravity of the violation, the economic benefit gained through non-compliance, and site compliance history into account. Additionally, the costs of compliance, the probability of detection, and the penalties associated with violations vary across states. Some states have regulations that are more stringent than federal regulations which obviously increases the costs of compliance. States also have different programs and policies in place that affect the probability that a facility will be inspected, and thus the probability of detection. Finally, because the RCRA penalty policy ultimately is just a guideline, the penalties associated with violations can vary across states.

This analysis uses data on facility characteristics and state regulations and policies to examine compliance behavior. The data are from 1995, the last year before EPA implemented a self-audit policy that significantly reduces penalties if a firm voluntarily discloses violations of environmental regulations.³ Prior to the self-audit policy, facilities had no monetary incentives to disclose violations and thus violations were only detected during the course of a facility inspection. If the facility was not inspected, there is no data on whether the facility was in compliance. To account for the censoring of the violation data, the analysis uses a censored bivariate probit similar to the type used in Greene (1992) and Stafford (2002).⁴ For simplicity, all violations are assumed to be willful, that is there are no inadvertent violations.⁵

Both the probability of a violation and the probability of an inspection are modeled as latent variables, $Y_{1i}^* = x_{1i}\beta_1 + \varepsilon_{1i}$ and $Y_{2i}^* = x_{2i}\beta_2 + \varepsilon_{2i}$, respectively. Let Y_{1i} and Y_{2i} be the observable binary variables associated with these two latent variables. Because regulators are more likely to inspect facilities that are suspected of being in violation, the error terms ε_{1i} and ε_{2i} should be positively correlated. Thus, the likelihood of observing a detected violation ($Y_{1i} = Y_{2i} = 1$) is:

$$L_{Y_{1i}=1, Y_{2i}=1} = \sum_{Y_{1i}=1, Y_{2i}=1} \log \Phi_2[x_{1i}\beta_1, x_{2i}\beta_2, \rho],$$

3 Provided that the firm corrects the violation in a timely manner (*Incentives for Self-Policing: Discovery, Disclosure, Correction and Prevention of Violations* (60 FR 66706)).

4 This method is similar to the “detection controlled estimation” model developed by Feinstein (1990), although it allows for the errors in the inspection and violation equations to be correlated.

5 Inadvertent violations will not affect the model as long as they are white noise. If firms are less likely to be fined if a violation is accidental, then the results of this analysis will be biased towards zero.

where Φ_2 is the bivariate normal cumulative distribution function and ρ is the covariance between ε_{1i} and ε_{2i} . The likelihood of inspecting a non-violator ($Y_{1i} = 0, Y_{2i} = 1$) is

$$L_{Y_{1i}=0, Y_{2i}=1} = \sum_{Y_{1i}=0, Y_{2i}=1} \log \Phi_2[-x_1\beta_1, x_2\beta_2, -\rho].$$

If a facility is not inspected, whether the facility is in violation is unknown. Thus the maximum likelihood function for the censored bivariate probit can be expressed as follows:

$$\begin{aligned} L = L_{Y_{1i}=0, Y_{2i}=1} + L_{Y_{1i}=1, Y_{2i}=1} + L_{Y_{2i}=0} = & \sum_{Y_{1i}=1, Y_{2i}=1} \log \Phi_2[x_1\beta_1, x_2\beta_2, \rho] \\ & + \sum_{Y_{1i}=0, Y_{2i}=1} \log \Phi_2[-x_1\beta_1, x_2\beta_2, -\rho] \\ & + \sum_{Y_{2i}=0} \log(1 - \Phi[x_2\beta_2]). \end{aligned}$$

3. Data and Results

Not all of the facilities subject to RCRA in 1995 are included in the analysis. Only facilities that have been consistently regulated under RCRA are included.⁶ This eliminates newly regulated facilities that might violate due to lack of knowledge about the regulations as well as one-time or infrequent generators. Additionally, federal facilities are excluded from the analysis, as the impact of penalties on federal facilities is not the same as the impact on for-profit companies. Also, small quantity generators are excluded because the regulations governing small quantity generators are significantly more lenient than those governing large quantity generators.⁷ The 8,216 facilities included in the analysis represent approximately 40% of the universe of large quantity generators and treatment, storage, and disposal facilities and account for almost 90% of hazardous waste managed in 1995.

As discussed in section 2, the two dependent variables in the analysis, Inspection and Violation, are binary variables that indicate whether the facility was inspected or found in violation, respectively.⁸ For this analysis, Inspection is equal to 1 if any state or regional

6 More specifically, only facilities that were regulated under RCRA for at least four years prior to 1995 are included in the analysis.

7 A facility is a small quantity generator if it does not generate more than 1,000 kilograms of hazardous waste in any single month.

8 The Inspection and Violation variables are derived from enforcement and compliance data recorded in EPA's RCRA Information System (RCRIS) which can be accessed via the Envirofacts database on EPA's website, www.epa.gov.

Table 1. Results of the Censored Bivariate Probit				
Variable	All Violations		Class 1 Violations	
	Inspection	Violation	Inspection	Violation
Constant	− 0.85* (0.19)	− 1.05* (0.23)	− 0.83* (0.19)	− 1.28* (0.24)
Permit	0.82* (0.06)	0.48* (0.06)	0.81* (0.06)	0.53* (0.06)
Generated	0.04* (0.01)	0.04* (0.01)	0.04* (0.01)	0.04* (0.01)
Received	0.03* (0.01)	0.02* (0.01)	0.03* (0.01)	0.02* (0.01)
Combustion	0.52* (0.15)	0.45* (0.11)	0.56* (0.15)	0.41* (0.11)
Land	0.37* (0.11)	0.13 (0.10)	0.37* (0.11)	0.06 (0.11)
Petroleum	0.54* (0.13)	0.27* (0.12)	0.51* (0.13)	0.37* (0.13)
Metal	0.17* (0.06)	0.17* (0.07)	0.18* (0.06)	0.21* (0.07)
Fabricated	− 0.11* (0.04)	− 0.09** (0.06)	− 0.10* (0.04)	− 0.07 (0.06)
Utilities	0.07 (0.09)	− 0.17** (0.10)	0.08 (0.09)	− 0.04 (0.11)
Age < 10	1.64* (0.50)	0.37 (0.25)	1.58* (0.50)	0.34 (0.26)
Age 10–20	0.87* (0.15)	0.36* (0.13)	0.90* (0.15)	0.47* (0.14)
Age 20–30	0.43* (0.05)	0.33* (0.06)	0.44* (0.05)	0.30* (0.07)
Age 30–40	0.43* (0.07)	0.25* (0.07)	0.43* (0.07)	0.18* (0.08)
Age 40–50	0.45* (0.08)	0.27* (0.08)	0.44* (0.08)	0.15 (0.09)
Age > 50	0.49* (0.06)	0.35* (0.07)	0.50* (0.06)	0.29* (0.07)
Inspected $_{-1}$	0.10* (0.04)	− 0.27* (0.06)	0.09* (0.04)	− 0.29* (0.06)
Violated $_{-1}$	0.16* (0.05)	0.25* (0.05)	0.16* (0.05)	0.23* (0.06)
State Inspections	0.54* (0.07)	0.04 (0.07)	0.53* (0.07)	0.06 (0.07)
State Violations $_{-1}$	− 0.41* (0.06)		− 0.57* (0.06)	
Strict	0.17* (0.05)	0.27* (0.07)	0.16* (0.05)	− 0.01 (0.06)
Voluntary P2		− 0.37* (0.07)		− 0.09 (0.07)
Mandatory P2		− 0.28 (0.42)		− 0.34 (0.51)
Indicators	− 0.13** (0.07)	0.04 (0.09)	− 0.07 (0.07)	0.05 (0.10)
Sustainable	− 0.42* (0.08)	− 0.38* (0.10)	− 0.44* (0.08)	− 0.63* (0.10)
Spending $_{-1}$	− 0.25* (0.11)	0.32* (0.15)	− 0.16 (0.11)	− 0.54* (0.02)
Employees	0.22* (0.05)	0.10 (0.07)	0.21* (0.05)	0.37* (0.07)
Headquarters	− 0.08 (0.12)	0.98* (0.19)	− 0.01 (0.12)	0.43* (0.19)
Costs $_{-1}$	− 0.17* (0.09)	− 0.27* (0.10)	− 0.20* (0.09)	0.02 (0.11)
ρ	0.92* (0.05)		0.94* (0.03)	
*Statistically significant at 5%; **Statistically significant at 10%; standard errors in parentheses. Regional dummies are not reported.				

official inspects the facility, regardless of the purpose of the inspection.⁹ The Violation variable has two specifications. For the first specification, Violation is equal to 1 if there is any RCRA violation at the facility, regardless of the severity of the violation. For the second specification, Violation is equal to 1 if there is any Class 1 violation at the facility. Class 1 violations are violations that (1) could result in hazardous waste being treated, stored, or disposed at an unauthorized facility, (2) could result in hazardous waste or

9 Inspections may be conducted to monitor compliance with all regulations, to evaluate specific systems such as groundwater monitoring devices, to review records, or to verify compliance with enforcement orders.

constituents being released to the environment, or (3) could inhibit the early detection of such releases.¹⁰ Violations that do not fall into Class 1 include primarily paperwork and recordkeeping violations. In 1995, almost 40% of the facilities in this study were inspected. Approximately half of those facilities (20% of all facilities) had at least one violation and almost one-third (12% of all facilities) had at least one Class 1 violation.

Table 1 presents the results of the censored bivariate probit regression for the two specifications of the Violation variable: All Violations and Class 1 Violations.¹¹ The inspection and violation equations contain both facility-specific and state-specific explanatory variables. The motivation for including each of these variables in the analysis and a discussion of the results for the two violation specifications follow.

3.1. Facility-Specific Variables

A number of variables characterizing the level and nature of hazardous waste activity at each facility were extracted from EPA's Biennial Reporting System (BRS).¹² Because there is more potential for environmental damages at facilities that manage large quantities of waste or conduct certain types of management operations, enforcement authorities are likely to target facilities based on hazardous waste activity. Additionally, Garvie and Keeler (1994) show that when regulatory resources are limited, firms with high compliance costs should attract more regulatory effort than lower cost firms. Although such targeting increases the probability that a facility will be inspected, facilities with more waste activity and thus higher compliance costs receive higher benefits from non-compliance. To capture differences in waste activity and compliance costs, the variables Permit, Generated, Received, Combustion, and Land are included in both the inspection and violation equations.

Permit indicates whether the facility has a RCRA operating permit. Most facilities that treat or dispose of hazardous waste on site must obtain a RCRA operating permit and permitted facilities are subject to more regulations than non-permitted facilities. Generated and Received measure the quantity of hazardous waste generated at the facility and the quantity received at the facility for management, respectively. Combustion indicates whether the facility operates a hazardous waste combustion unit (e.g., an incinerator or industrial furnace) and Land indicates whether the facility operated a hazardous waste land disposal unit (e.g., landfill or surface impoundment). Except for the coefficient on Land in the violation equation, the coefficients on these variables are positive and significant in both equations for both specifications. This suggests that the increased costs of compliance for permitted facilities, facilities that generate or receive large quantities of waste, and facilities with combustion units outweigh the increased

10 The Class 1 definition may be expanded by a state with the permission of EPA. Also class determination is made by the inspector who discovers the violation.

11 If a non-Class 1 violation is discovered at the same time as a Class 1 violation, it may be recorded as only a Class 1 violation. This makes it difficult to obtain an accurate count of facilities with non-Class 1 violations, so I chose not to include a non-Class 1 specification.

12 The BRS also can be accessed from the Envirofacts database on EPA's website, www.epa.gov.

probability that the facility will be inspected.¹³ For facilities with land disposal units, the increased probability of inspection may outweigh additional compliance costs.

Petroleum, Metal, Fabricated, and Utilities are dummy variables indicating a facility's primary industrial classification. As Chua et al. (1992) demonstrate, firm compliance will depend on industry structure, particularly the size of the regulated industry. Thus, it is likely that inspection and violations probabilities vary across industries. Additional industry dummies were included in preliminary analyses, but only these four industry classifications were significant and are thus included in the analysis reported here. For the industry dummies, the results suggest that facilities in petroleum/coal products and primary metal industries are both more likely to be inspected and more likely to violate the regulations. Facilities in the fabricated metals industry are less likely to be inspected and less likely to violate the regulations. These results are consistent with the Chua et al. (1992) model as the petroleum/coal products and primary metal industries are both relatively small while the fabricated metal industry is relatively large.¹⁴ Utilities do not appear to be more or less likely to be inspected (the coefficients in the inspection equations are not significant) but may be less likely to violate regulations given the negative and significant coefficient in the All Violations specification.

Age dummies are included in the analysis because age should affect the cost of compliance and thus the benefit from violation as well as the probability of inspection. Although age was reported for only 30% of the facilities in the analysis, the age dummies are all positive and are almost all significant in both equations for both specifications. Thus compared to facilities with no age reported, facilities whose age was reported are more likely to be inspected and more likely to violate the regulations. Using additional analyzes, it can also be shown that facilities that are less than 20 years old are more likely to be inspected than facilities that are more than 20 years old, although there is no statistically significant difference in the likelihood of violation. Although age could affect inspection and violation probabilities through differences in capital vintages, another possible explanation for these results is that facilities that are inspected more often are better known to regulators and thus additional information on these facilities, like facility age, has been recorded.

The final two facility-level variables, Inspected₋₁ and Violated₋₁, indicate whether the facility was inspected or found in violation in the previous year, respectively. Inspected₋₁ is included in the analysis because according to RCRA policy every facility should be inspected at least once every two years. If the facility was inspected in the previous year, it is less likely to be inspected this year, *ceteris paribus*. If a facility is found in violation in the previous year, EPA is more likely to conduct a follow up inspection this year to ensure

13 The results for Permit, Generated, and Received are consistent with Stafford (2002). Combustion and Land were not included in that analysis.

14 According to the 1997 Economic Census there were approximately 2,000 establishments in the petroleum/coal products industry, 5,000 in the primary metals industry, 15,000 in the utilities industry, and 62,000 in the fabricated metals industry.

that past offenders have come into compliance with the regulations.¹⁵ Also, penalties can be increased if the facility has a past history of violations.¹⁶

As is generally true for the other facility-specific variables, there is no qualitative difference in the results for Inspected_{-1} and Violated_{-1} between the two specifications. The coefficient on Inspected_{-1} is positive and significant in the inspection equation, indicating that firms that were inspected in 1994 were more likely to be inspected in 1995. This suggests that there may be an unobserved or omitted characteristic of the facility that the enforcement agency is targeting. For example, there may be specific activities or substances at the facility that make the facility more likely to be inspected, or the facility could be conveniently located (close to a regional office or other regulated facilities). The coefficient on Inspected_{-1} in the violation equation is negative and significant, perhaps because facilities that were inspected in 1994 believed (correctly) that they were more likely to be inspected and thus if they violated were more likely to be caught.

Facilities that violated in 1994 also were more likely to be inspected in 1995, as expected. Harrington (1988) shows that this is the optimal enforcement strategy when penalties are restricted. The coefficient on Violated_{-1} in the violation equation is positive and significant, which seems to be inconsistent with the higher probability of an inspection for facilities that violated in 1994 and the fact that penalties are higher for repeat violators. However, this is consistent with Harrington's model in which there is a class of high-cost firms that will always be in violation. There may be an unobserved or omitted variable such as compliance cost or corporate culture that make a facility more or less likely to violate. Also, it may take time to correct a violation, particularly if it requires installing new technology, so that past violations may in part determine current violations.

3.2. State-Level Variables

The state-level data come from a variety of sources and capture different aspects of a state's hazardous waste program. *Ex ante* it is not obvious whether a specific state regulation or policy will increase or decrease compliance. On one hand, a regulation or policy may serve to increase the costs of compliance, and thus increase the benefit to a facility of non-compliance. On the other hand, if that policy also increases the expected cost of non-compliance through an increased likelihood of being caught or higher penalties once caught, we could see a higher level of compliance.¹⁷ In addition, a regulation may appear to be relatively stringent as written, but if implementation or enforcement is weak, it may not affect compliance behavior. The purpose of this analysis

¹⁵ For both specifications, Violated_{-1} is equal to 1 if there was any violation in the past year.

¹⁶ In both civil and criminal cases it is common practice to increase punishment for repeat offenders. Polinsky and Rubinfeld (1991) model the conditions under which increasing fines for repeat offenses will be optimal.

¹⁷ Although in general there are no theoretical models that explicitly model the policies included in the analysis, there are a number of theoretical papers with this same underlying intuition. For example, in Kambhu (1989) increases in regulatory standards can result in increased non-compliance if engaging in penalty reducing activities becomes less expensive than complying with the more stringent standard. Similarly in Nowell and Shogren (1994) increasing penalties increases the incentives to challenge enforcement efforts and in some cases may result in lower levels of compliance.

is to examine which regulations and policies are most effective in promoting facility compliance.

State Inspections is the number of inspections (in hundreds) conducted in the state, normalized by the gross product of the state.¹⁸ The coefficient on State Inspections has the expected positive and significant sign in the inspection equation for both specifications, as a higher level of inspection activity increases the probability that a given facility will be inspected. The coefficient in the violation equation is also positive, although not significant in either specification.

State Violations₋₁ measures the number of violations (in hundreds) detected in the state in the previous year, normalized by the gross product of the state.¹⁹ This variable is included because if there are many violations in the previous year, the state might devote more resources to conducting follow-up inspections, and thus fewer general inspections will be made at any one facility. As discussed in Poirier (1980), the bivariate probit model will be identified if there is at least one variable that affects whether the facility will be inspected but does not affect whether the facility violates the regulations, as well as at least one variable that affects whether the facility violates the regulations but does not affect whether the facility will be inspected. State Violations₋₁ is included in the inspection equation but is not included in the violation equation. Although the number of violations in the state in a previous year should affect the probability that an inspection will be conducted in the current year, data on the level of violations will not generally be common knowledge to facilities and therefore should not affect the facility's violation decision.²⁰ The coefficient on State Violations₋₁ is negative, as expected, and significant in the inspection equation for both specifications.

The next set of variables, Strict, Voluntary P2, Mandatory P2, Indicators, and Sustainable all indicate various regulations or policies that states may have in addition to federally-mandated regulations. Strict is equal to 1 if the state has a strict liability regime.²¹ Under strict liability, firms are responsible for any damages caused by releases of hazardous waste into the environment regardless of whether the firm acted negligently or with intent. In terms of enforcement behavior, strict liability regimes may give enforcement agencies stronger incentives to find the potentially responsible parties associated with a spill or accident, and thus stronger incentives to inspect facilities, because damages can be recovered from such parties without proof of negligence or intent. The expected effect of strict liability on firm compliance is not obvious. Although one

18 State Inspections includes all inspections conducted in the state according to RCRIS, not just those conducted at facilities in this analysis. Gross State Product data was obtained from the Bureau of Economic Analysis.

19 State Violations₋₁ includes all violations found in the state according to RCRIS, not just those found at facilities in this analysis.

20 This information is publicly available in RCRIS, although there generally is a lag between violations being detected and entered into RCRIS. Also, EPA's annual report on RCRA Enforcement does not provide state level data.

21 U.S. EPA (1995) provides data on state liability regimes.

might expect strict liability to increase the cost of violations, Beard (1990) shows that a strict liability regime may result in either too much or too little care.²²

First, consider the results for the inspection equation. For both specifications, the coefficient on Strict is positive and significant, as expected.²³ For the violation equation, however, the results depend on the specification. In the All Violations specification, the coefficient is positive and significant while in the Class 1 specification, the coefficient is negative but not significant. Although the positive coefficient in the All Violations specification is the opposite of what was expected, it is consistent with Alberini and Austin's (1999) finding that there is a positive correlation between spills and strict liability regimes. Further analysis by Alberini and Austin (2002) concludes that factors that increase the likelihood of adopting strict liability, such as high numbers of manufacturing and small firms, are also positively correlated with spills. This explanation is likely to hold for violations as well because many spills also qualify as RCRA violations. The difference between the coefficients on Strict in the violation equation for the two specifications is also interesting. If strict liability has any deterrent effect on violations, one would expect it to be more pronounced for Class 1 violations, since those violations are more likely to result in damages. For the Class 1 specification, the negative and insignificant coefficient suggests that any positive correlation between adoption of strict liability and violation probability is offset by increased incentives for compliance.

Voluntary P2 and Mandatory P2 indicate whether the state has a voluntary or mandatory pollution prevention (P2) program, respectively.²⁴ Pollution prevention, also known as source reduction, is the preferred "management" method for hazardous waste because waste is eliminated rather than treated or contained. By 1995, over half of the states had passed laws generally confirming that pollution prevention was a good idea and encouraging voluntary pollution prevention efforts. Two states, Arizona and California, had mandatory pollution prevention programs (e.g., reduction of waste by 5% by a certain date). Because there is no obvious relationship between pollution prevention programs and state inspection policies, these variables are the explanatory variables omitted from the inspection equation.²⁵ In the violation equation, the coefficients on the P2 variables are all negative, although only the coefficient on Voluntary P2 in the All Violation specification is significant. Because the primary focus of many voluntary pollution prevention programs is facility education and guidance, such efforts may be able to influence facility compliance with respect to paperwork and "good housekeeping" practices.²⁶ However, such efforts

22 A number of papers in addition to Beard (1990) have looked at the effect of different liability rules on compliance (for example, Shavell 1984 and Ringleb and Wiggins 1990). As in Beard, the effect of strict liability on compliance depends on the characteristics of the firm and/or the penalty structure.

23 However, this result is inconsistent with Stafford (2001) which finds that during 1986 to 1995, the adoption of strict liability was associated with a decrease in the probability of inspection.

24 Data on pollution prevention programs was obtained from the National Pollution Prevention Roundtable (1999).

25 Moreover, when these variables were included in the probit regression of the Inspection equation, they were not significant.

26 Although there are a few theoretical papers on voluntary environmental programs, these papers primarily focus on how voluntary programs compare to mandatory programs (for example, Arora and

may be less effective at reducing other types of violations. Also, the insignificant coefficients on Mandatory P2 could be due to the fact that only two states had adopted mandatory pollution prevention laws by 1995.

The next two variables capture different aspects of state program planning and implementation.²⁷ Indicators reflects whether the state used environmental indicators in program planning or implementation. Environmental indicators are measures of environmental quality used to determine success in implementation of environmental programs. Environmental indicators measure program performance (e.g., human exposures to hazardous constituents) rather than program activity (e.g., number of violations detected), the traditional method for evaluating environmental programs. Since environmental indicators are more flexible than traditional performance measures, they might encourage compliance. Sustainable indicates whether sustainable development is a priority in program planning or implementation.²⁸ Because sustainable development is often at odds with a more traditional view of economic development, making sustainable development a priority could indicate the importance of the environment in a state.

The results of both specifications suggest that the use of environmental indicators has no significant effect on compliance. Moreover, the coefficients are positive rather than negative as expected. However, states where sustainable development is a priority do appear to have higher compliance rates. In terms of the inspection equation, states with sustainable development priorities have a significantly lower probability of inspections, although it is not obvious why that should affect inspection levels. It could be that the movement towards sustainable development is supported or spearheaded by business leaders and citizens who have a high regard for the environment that is also reflected in higher compliance rates.²⁹

The next set of variables, *Spending₋₁*, *Employees*, and *Headquarters* measure state resources devoted to the environment. The data were collected by the Council of State Governments (1996) by examining legislative and executive budget documents. *Spending₋₁* measures state spending (in thousands) on hazardous and solid waste in the previous year, normalized by state gross product. Because many states do not split hazardous waste spending out from solid waste spending (since hazardous waste is a subset of solid waste), total waste spending is used in the analysis. Data from the previous year are used because the relationship between spending in a given year and the number of inspections and violations in that year could be endogenous while lagged spending should not be. *Employees* measures the number of employees in the state environmental agency

Gangopadhyay 1995 and Segerson and Miceli 1998). The existing empirical literature on voluntary programs focuses on the determinants of firm participation (Arora and Cason 1995 and Videras and Alberini 2000).

27 The Council of State Governments (1994) compiled the data for these variables.

28 Sustainable development is admittedly a vague term and can encompass many different policies. This variable captures whether states list "sustainable development," however defined, as a priority.

29 Including additional demographic variables in the analysis such as the percentage of citizens that are members of environmental organizations or whether the state governor is a Republican does not qualitatively change these results.

and Headquarters measures the percentage of employees located in the headquarters office as opposed to regional or local offices.

First, consider the results for the violation equation. The coefficient on Spending_{-1} in the violation equation is positive and significant in the All Violations specification, but negative and significant in the Class 1 Violations specification. Thus higher spending levels do deter Class 1 violations, but are positively correlated with violations in general. The negative and significant coefficient on Spending_{-1} in the inspection equation suggest that there may be more targeting in states with higher spending. If the states target facilities that are most likely to commit Class 1 violations, we might expect to see a decrease in Class 1 violations coupled with an increase in non-Class 1 violations. This result would also be consistent with a “filtered” enforcement regime like that of Heyes (2002), where more intensive and repeated inspections are conducted at a smaller number of facilities that are most likely to be serious violators.

The coefficient on Employees is positive in both equations for both specifications, and significant in three of the four cases. Thus the more employees, the higher the probability of an inspection. The positive correlation between Employees and Violations probably reflects the need for more employees in states with the potential for and past history of significant violations. The coefficient on Headquarters is not significant in the inspection equation for either specification, but is positive and significant in the violation equation for both specifications. This suggests that locating employees in regional offices can increase compliance, perhaps by establishing better working relationships between the regulators and the regulated community.

The final state variable, Costs_{-1} , is an index of average pollution abatement costs across all media in the previous year. Levinson (2001) calculated this index by adjusting survey data on pollution abatement costs and expenditures (PACE) to account for state differences in industrial composition. Abatement costs can be used to measure the stringency of state environmental regulation as more stringent regulation would result in higher pollution abatement costs. Of course this measure will also capture other state differences such as differences in wages. As with Spending_{-1} , the lagged value is used because abatement costs should depend on the level of violation in a given year while lagged abatement costs should not. For both specifications, Costs_{-1} is negatively correlated with the probability of an inspection. This suggests that to some extent regulatory stringency and enforcement may be substitutes, not complements. For the All Violations specification, the coefficient on Costs_{-1} is negative and significant in the violation equation as well, which is the opposite of what was expected, i.e., more stringent regulations provide more incentives to violate. It could be that the higher costs reported in the PACE survey reflect higher levels of compliance in addition to more stringent regulations.

Regional dummies were also included in the regression, although to simplify presentation of the results, they are not included in table 1. While not all of the regional dummies are significant, they are jointly significant according to likelihood ratio tests. Finally, note that the correlation coefficient for the two equations, ρ , is positive as expected and significant. This suggests that there may be unobserved or unmeasured characteristics that make facilities more likely to violate the regulations and make a facility more likely to be inspected.

4. Conclusion

In general, the results of this analysis are consistent with earlier work on hazardous waste compliance with respect to the effect of facility-level variables on compliance behavior. This study also extends that work by analyzing more closely the relationship between different state policies and programs and facility compliance behavior. The results of this analysis provide weak evidence that state adoption of a strict liability regime may decrease Class 1 violations of RCRA regulations. The effect of strict liability on other violations is difficult to assess within the framework presented here because violation levels are likely to be positively correlated with the adoption of strict liability regimes. Adoption of voluntary pollution prevention programs, on the other hand, appears to discourage non-Class 1 violations, although it does not have an effect on more serious violations. Mandatory pollution prevention programs do not affect compliance, although it is hard to analyze their effect since only two states have adopted such programs. The use of environmental indicators in program planning also does not affect facility compliance, although states that incorporate sustainable development into program planning do have higher levels of compliance.

With respect to the effect of resources on compliance behavior, both in terms of spending and employees, the evidence is mixed. The results show a positive relationship between spending and violations in general, but a negative relationship between spending and Class 1 violations. The most likely explanation for this result is that higher resource levels reflect higher resource requirements due to a high level of noncompliance. However, higher spending levels do appear to decrease the number of Class 1 violations, perhaps through increased targeting. States also can increase compliance by allocating a higher percentage of employees to regional offices.

While the results of this analysis do increase significantly our understanding of the effectiveness of various policies on hazardous waste compliance, there are three important factors that this analysis is unable to address due to the lack of data. The first is the effect of monetary sanctions on compliance. Although EPA has developed guidance on the setting of monetary penalties, regulators have significant discretion in determining the amount of the fine that facilities will pay. Unfortunately, data on fines paid is not readily available. Additionally, it is the expected fine, rather than the actual fine that should effect compliance behavior. Second, the analysis does not examine the effect of criminal sanctions, either in addition to or instead of monetary fines, on compliance behavior although research in other areas suggests that the use of criminal sanctions varies across, and even within, jurisdictions.³⁰ Finally, the analysis does not consider the effect of regulatory dealing, that is a comprehensive agreement between a firm and the regulatory agency covering multiple media and environmental programs.³¹ Further research that incorporates these issues will help to improve our understanding of firm compliance and the ability of regulators to develop effective enforcement programs.

30 See, for example, Cory and Germani (2001).

31 Heyes and Rickman (1999) develop a model of regulatory dealing and how it would be incorporated in an optimal monitoring and enforcement regime.

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